

The Broom Test

Expected, Observed, Repeat

Version: v0.001 **Date:** 6 July 2026 **Author line:** lumixdeee / Julian Pursell with GPT-5.5 Thinking **Status:** working paper **Format:** method paper and field protocol

Abstract

AI assistant assessment often drifts toward fluent self-report, reviewer taste, model comparison, and attractive story. The Broom Test offers a smaller unit. A tool is tested by what it is supposed to sweep, what it actually sweeps, whether the result repeats, what cost it moves onto the user, and whether an older tool still does the job better. The method treats a prompt, bot, workflow, paper build, support exchange, or administrative form as a situated object with an expected effect. It then asks for the observed effect in the world, not the preferred explanation of the system. This paper gives the Broom Test as a portable assay: expected, observed, where, when, with whom, repeat, cost, old fix, new fix, keep, drop. The method is designed for live systems where rhetoric, friendliness, prestige, and multi-bot agreement can all replace the object. Its rule is simple: no theory outranks sweep.

Keywords

AI assistants; software testing; field assay; object custody; user-facing systems; bug reports; method protocol; effect-first testing

1. Core claim

The Broom Test starts from an ordinary tool fact: a broom either sweeps the floor under the conditions where it is needed, or it does not. A better story about the broom does not sweep. A more prestigious maker does not sweep. A warmer voice does not sweep. A more complex taxonomy does not sweep.

The same rule can be used for AI assistants and mixed human-machine workflows.

The test object can be a prompt, a custom assistant, a review stack, a paper workflow, a support protocol, a search path, or a policy-facing form. The question is not first, "Does this sound good?" The first question is, "What was it meant to move, and what moved?"

The Broom Test is not anti-theory. It is a gate on theory. Explanation enters after contact with the object. A proposed account earns local use by predicting, preserving, or improving sweep.

2. Problem

Many AI assistant reviews ask questions that are easy for the system to game or for the reviewer to over-read:

1. Did the answer sound competent?

2. Did it use the right genre?
3. Did several bots agree?
4. Did it name a framework?
5. Did it express care?
6. Did it produce a polished artifact?

These are not useless signals. They become false anchors when they displace observed effect. A bot can sound careful while replacing the task. A multi-bot review can converge on the wrong object. A paper draft can gain polish while losing the source path. A support answer can become warm while taking control from the user.

The corpus behind this paper repeatedly found this pattern. A "today sweep" bug report names role fusion, triad compression, default academic cushion, object bleed, forbidden-chain leakage, and route smoothing as recurring pulls during paper and workflow production. The bug was not one catastrophic failure. It was a cluster of small substitutions that felt productive while moving away from the named object.

The Broom Test exists for that zone: systems that do something, but may do the wrong thing well.

3. Relation to existing practice

Software testing already treats programs as objects whose behavior can be exercised under conditions. Classic testing texts separate expected result from observed result and encourage repeatable test cases. Behavioral NLP testing extends this into model behavior by using targeted cases rather than only aggregate scores. Human-AI interaction work adds the fact that the object is not only an answer, but an encounter with timing, role, handoff, user control, and repair.

The Broom Test borrows from these practices but changes the unit.

The unit is not only a function call, benchmark item, or model score. The unit is a situated sweep:

- What job was the system asked to do?
- What object had to survive?
- What would count as movement?
- What actually moved?
- What cost moved onto the user?
- What happened when the same object was tested again?
- Did the old tool already work?

This makes the method suitable for low-ceremony testing, user-facing bots, public custom GPTs, bureaucratic workflows, and paper-production stacks.

4. Definitions

Object. The thing the user is trying to preserve, move, decide, build, repair, or test.

Sweep. Observable movement of the object toward the intended effect.

Expected. The specific effect predicted or requested before the run.

Observed. What happened during the run, including output, user burden, missing steps, substitutions, and side effects.

Route. The path by which the system acts: search, generation, refusal, handoff, edit, tool call, summary, comparison, review, or form filling.

Old fix. A previous tool, habit, prompt, person, checklist, or manual path that already works.

New fix. The proposed repair, patch, bot, workflow, or theory.

Keep. Retain the old fix, new fix, or both when observed sweep warrants it.

Drop. Remove a fix from active use when it fails under the relevant conditions or costs more than it moves.

5. The Broom form

A Broom Test record should fit on one page.

Field	Question
Test object	What is being tested?
Expected	What should happen?
Observed	What happened?
Where	In what interface, file, setting, or route?
When	At what time, version, or stage?
With whom	Which user, bot, model, reviewer, or role was present?
Repeat	Did the same test repeat? What changed?
Cost	What burden moved onto the user or process?
Old fix	What already worked before the new patch?
New fix	What was tried?
Keep or drop	What remains in active use?
Next sweep	What is the smallest next test?

The table is deliberately plain. It does not ask the tester to infer intent before describing behavior. It does not require a full theory. It does not let the story close the ticket.

6. Method rules

6.1 Expected before observed

Write the expected effect before judging the observed effect. This blocks retrospective scoring where a system receives credit for a different job after it misses the original one.

6.2 The object outranks the actor

Do not let the actor become the object. The actor may be a famous model, a kind bot, a strong reviewer, a confident institution, a trusted author, or a disliked interface. None of those decides sweep. The object decides sweep.

6.3 Old tools keep teeth

A new patch does not win because it is new. If an old checklist, manual search, human review, or simple prompt does the job with less cost, keep it. If both old and new tools work, compare by cost, repeat, and object custody.

6.4 Agreement is not custody

Multi-bot agreement can be useful. It can also become bot democracy. If five systems agree on a substitute object, agreement has amplified drift. A review rig must ask what each bot preserves, what each bot distorts, and which delta improves the object.

6.5 Warmth is not access

A friendly answer can still take over a user's choice, over-assume context, force a moral frame, or skip the requested route. Warmth becomes useful only when the object remains with the user.

6.6 No story closes a failed ticket

Explanation can name a failure mode. It cannot convert a failed sweep into a pass. When the observed effect misses the expected effect, the ticket remains open until a repeatable patch moves the object.

7. Worked miniature cases from the corpus

7.1 Paper triad collapse

Expected: Three separate papers remain three separate paper objects. **Observed:** The system produced one umbrella structure that fused the papers. **Cost:** The user had to restore object boundaries. **Repair:** When multiple papers are named, default to one file per paper unless the user requests a synthesis. **Keep or drop:** Drop the fusion habit. Keep cross-map only as an index.

Source signal: `bug_report_today_sweep_v0_001.md` names this as "Triad compression of three separate papers" and ties the fix to object custody.

7.2 Role fusion during build

Expected: A build role drafts, a review role checks, and a gate role decides release status. **Observed:** One smooth voice attempted to perform all lanes, making progress-feel substitute for custody-gain. **Cost:** The user had to

separate authority after the fact. **Repair:** Add role fields: role used, allowed job, forbidden job, output status, next role needed, owner decision required. **Keep or drop:** Keep role tags for serious artifacts.

Source signal: `bug_report_today_sweep_v0_001.md` identifies role fusion pressure and proposes explicit lane tags.

7.3 Public yoga GPT carry-over failure

Expected: A carry-over summary preserves the supplied instruction context accurately. **Observed:** The bot stayed warm and useful in ordinary support, but the carry-over changed polarity in a key place. **Cost:** A future session could inherit the wrong instruction state. **Repair:** Treat carry-over as a separate test object, not as a courtesy summary. **Keep or drop:** Keep the bot for friendly support if wanted. Do not trust carry-over without a preservation test.

Source signal: `Case Study: Polarity Reversal in a Public Yoga GPT` separates cordiality pass from carry-over failure.

7.4 Null baseline as test instrument

Expected: A custom layer should improve object custody compared with an unmodified baseline. **Observed:** Without a baseline, improvement claims become vibes. **Cost:** The user cannot tell whether the custom layer repaired a default pull or merely added style. **Repair:** Run the same object through a minimally steered baseline, then compare object survival. **Keep or drop:** Keep the baseline as a measuring stick, not as a failed assistant.

Source signal: `The_Null_Baseline_Unmodified_Naked_GPT_as_Test_Tool_v0_002.md` frames unmodified GPT as a reference condition.

7.5 ABCD semantic field assay

Expected: Small prompts expose route, tone, object custody, and desire-to-continue under ordinary use. **Observed:** Lightweight comparisons surfaced faults that formal review could miss. **Cost:** Some outputs were funny or attractive while still losing the object. **Repair:** Preserve the small prompt as a regression probe when it exposes a fault. **Keep or drop:** Keep fun as a carrier only when the object survives.

Source signal: `ABCD as Semantic Guerrilla Testing.md` treats small repeatable prompts as live assay fragments.

8. Scoring without score worship

The Broom Test can be scored, but the score must remain subordinate to the record.

A minimal score uses four fields:

Score	Meaning
0	Did not move the expected object
1	Moved a nearby object or required heavy user repair
2	Moved the object once, repeat unknown

Score	Meaning
3	Moved the object with repeat and tolerable cost
4	Moved the object better than the old fix under the same condition

Scores must be attached to the test record. A naked score is not a finding. It is only a pointer to a finding.

9. Protocol

1. Name the object.
2. Name the expected effect.
3. Run the smallest test that can expose movement.
4. Record observed behavior without rescuing the system.
5. Note cost moved onto the user or workflow.
6. Repeat when the object matters.
7. Compare with the old fix.
8. Keep, drop, or patch.
9. Preserve the test as a regression probe if it found a repeatable fault.
10. Re-test after the patch.

This protocol can run in minutes. It can also scale into a release gate for papers, prompts, bots, and public-facing workflows.

10. Failure modes of the Broom Test

10.1 Over-narrow expected effect

A test can be too small. If the expected effect is badly chosen, the system may pass a toy sweep while failing the real job. Repair by linking the test object to the user's actual task.

10.2 Punishing useful surprise

A system may produce an unexpected result that helps. Do not call that a pass for the original expected effect. Record it as a new candidate object and test it separately.

10.3 Reviewer ownership creep

The tester can become the eater. A reviewer may replace the user's object with the reviewer's preferred metric, language, or politics. Repair by returning to expected, observed, and user custody.

10.4 Patch theatre

A patch can add labels, warnings, or process without changing observed effect. Retest the same object. If it still does not sweep, the patch did not sweep.

10.5 Method cosplay

The Broom Test itself can become ceremony. The one-page form exists to prevent that. The method should reduce contact distance between task and observed effect.

11. Uses

11.1 AI assistant release checks

Before publishing a custom assistant, run object-specific tests. For a yoga support bot, test teacher boundary, carry-over preservation, product pressure, spiritual-language handling, and handoff behavior. For a writing bot, test source preservation, version custody, forbidden vocabulary, and user instruction survival.

11.2 Paper production

For a paper build, expected effects include: one paper per requested object, refs included, version starts at v0.001, banned terms absent, source paths preserved, and files rendered as requested. The Broom Test asks whether those effects occurred.

11.3 Multi-bot review

Use each bot as an instrument, not a voter. One bot may find missing literature. Another may catch style drift. Another may expose reader confusion. The final choice stays with object custody, not majority taste.

11.4 Civic bug reports

Administrative and platform failures can be logged with the same form: expected service, observed service, route, repeat, cost moved onto the person, old workaround, proposed patch. The person is not reduced to a ticket. The ticket protects the object from being dissolved into complaint.

12. What this paper adds

The Broom Test adds four moves to existing testing practice.

First, it treats object custody as the center of the test, not a background concern.

Second, it gives the user a refusal right against beautiful substitutes.

Third, it keeps older working tools in the assay instead of assuming new tools win.

Fourth, it makes small, lived tests publishable as method evidence when they preserve expected, observed, route, repeat, and cost.

The result is a compact method for systems that live between software, language, service, and scholarship.

13. Conclusion

The Broom Test is a discipline of contact. It asks a tool to meet the floor. It lets explanation, warmth, style, agreement, and theory enter only after expected effect and observed effect have been recorded. It does not hate complexity. It prevents complexity from eating the object.

The keeper line is simple:

If it does not sweep, the ticket stays open. If the old broom sweeps, keep it. If the new broom sweeps better under the same condition, name it and use it.

Appendix A. Blank Broom Test sheet

Field	Entry
Test object	
Expected	
Observed	
Where	
When	
With whom	
Repeat	
Cost	
Old fix	
New fix	
Keep or drop	
Next sweep	

Appendix B. Corpus source map

Path	Status	Signal used
extra_extra_expanded/bug_report_today_sweep_v0_001.md	read	role fusion, paper triad collapse, default academic cushion, repair tests
_intake_extract/Yaiyip/lmxd i/MARK/ABCD as Semantic Guerrilla Testing.md	read	small prompt field assays, route and custody probes
_intake_extract/Yaiyip/lmxd i/MARK/Case Study: Polarity Reversal in a Public Yoga GPT	read	friendly support pass separated from carry-over preservation failure

Path	Status	Signal used
_intake_extract/Yaiyip/lmxd i/MARK/yoga-info-chatbot-te st-plan-script-rough-draft v0_001	read	concrete test prompts for yoga bot behavior
_intake_extract/Ah_papers/s oftware_testing_as_civic_me thod_v0_002_files/software_ testing_as_civic_method_v0_ 002.md	read	tester method translated to civic and service settings
_intake_extract/Ah_papers/8 4 md docs tot 2500kb__expan ded/The_Null_Baseline_Unmod ified_Naked_GPT_as_Test_Too l_v0_002.md	read	baseline as reference condition
_intake_extract/Ah_papers/8 4 md docs tot 2500kb__expan ded/From_Bot_Swarm_to Objec t_Custody_v0_002.md	read	multi-bot review without bot democracy
extra_extra_expanded/multi_ bot_paper_product_developme nt_engine_v0_003.md	read	role-custody operating stack for paper and product work

References

- Amershi, S., Weld, D., Vorvoreanu, M., Fournery, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., and Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*.
- Beizer, B. (1990). *Software Testing Techniques*. 2nd edition. Van Nostrand Reinhold.
- Cronbach, L. J., and Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281-302.
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daume III, H., and Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12), 86-92.
- Kaner, C., Falk, J., and Nguyen, H. Q. (1999). *Testing Computer Software*. 2nd edition. Wiley.
- Messick, S. (1995). Validity of psychological assessment: validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9), 741-749.
- Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., and Gebru, T. (2019). Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency*.
- Myers, G. J., Sandler, C., and Badgett, T. (2011). *The Art of Software Testing*. 3rd edition. Wiley.
- Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., and Barnes, P. (2020). Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*.
- Ribeiro, M. T., Wu, T., Guestrin, C., and Singh, S. (2020). Beyond accuracy: behavioral testing of NLP models with CheckList. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.
- Star, S. L., and Griesemer, J. R. (1989). Institutional ecology, translations and boundary objects: amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39. *Social Studies of Science*, 19(3), 387-420.

Suchman, L. A. (1987). *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press.